

Vitellaria paradoxa Shea Butter Tree PFAF Plant Database

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Translate this page:

Summary

Vitellaria paradoxa, or commonly known as Shea Butter Tree is a deciduous tree with a spreading crown and grows about 25 m in height. It is indigenous to Africa and is the sole species in the *Vitellaria* genus. Its bark is corky. The leaves are oblong and clustered at the ends of branches. The flowers are white and clustered at the ends of shoots. The fruits are flat and round containing up to four shiny brown seeds. Fruiting commence 10-15 years after planting but full production occurs at 20-30 years. Medicinally, Shea Butter is used for topical medicines against rheumatic and joint pains, wounds, swellings, dermatitis, bruises, and other skin conditions. It is also useful as relief from nasal congestion and rhinitis. The leaves are used to treat stomach pain and headache, and as eye bath. Ground roots and bark are used to treat diarrhea, jaundice, and stomach ache. Bark infusions have antimicrobial properties and are used against dysentery. Bark decoction, on the other hand, are used in baths to facilitate childbirth and stimulate lactation among feeding mothers. Seed kernel contains fat, also called as shea butter, which is white, odorless, and of high quality. It is used for cooking, pastries, and confectionery, and as an excellent substitute for cocoa butter. It is also used in cosmetic products, soap, and candles. Mature fruits are eaten fresh and flowers are made into fritters. The bark yields reddish latex which is used as a chewing gum, or made into glue and balls as toys. In addition to the above-mentioned uses, the tree is also planted for soil and water conservation purposes as well as for environmental protection. The wood is moderately heavy and resistant to termites. It is used for poles, house posts, rafters, flooring, utensils, and furniture. It is also an excellent fuel wood and can be made into charcoal.

Physical Characteristics

Vitellaria paradoxa is a deciduous Tree growing to 15 m (49ft) by 15 m (49ft) at a slow rate.

See above for USDA hardiness. It is hardy to UK zone 10. The flowers are pollinated by Bees. The plant is self-fertile.

Suitable for: light (sandy), medium (loamy) and heavy (clay) soils, prefers well-drained soil and can grow in nutritionally poor soil. Suitable pH: mildly acid, neutral and basic (mildly alkaline)



soils. It can grow in semi-shade (light woodland) or no shade. It prefers dry or moist soil.

[UK Hardiness Map](#) [US Hardiness Map](#)



Synonyms

Bassia parkii G.Don Butyrospermum mangifolium (Pierre ex A.Chev.) A.Chev.
Butyrospermum niloticum Ko

Habitats

Edible Uses

Edible Parts: FlowersFruitInner barkOilSapSeed

Edible Uses: GumOil

The kernel of the seed (often incorrectly called 'nut') contains a vegetable fat known as shea butter[63 , 299 , 301]. Shea butter from fresh seeds is white, odourless and of high quality, while that from stale seeds is dark, and tastes bitter[299]. High quality shea butter is consumed throughout West Africa as a cooking fat. Refined fat has been marketed as margarine and baking fat. It is used for pastries and confectionery because it makes the dough pliable[63 , 299]. It is a useful cocoa butter substitute because it has a similar melting point (32 - 45°C) and high amounts of di-stearin (30%) and some stearo-palmitine (6.5%) which make it blend with cocoa butter without altering flow properties[299]. In rural areas, seeds are traditionally processed by hot water extraction, usually the job of women. The fruit pulp is first removed for food, or by fermentation or boiling. The seeds are then boiled and later sun- or kiln-dried. Sun-drying may take 5 - 10 days. Seeds are cracked using mortar and pestle, or stones; the kernels are removed by trampling and redried before being crushed, ground and kneaded to form a paste; the paste is put in water, heated or boiled and the boiling mass is churned until a grey, oily fat separates from the emulsion. The fat is skimmed off from the surface and washed to remove impurities. The congealed fat may subsequently undergo further refining before being moulded in to various forms[299]. This traditional method of processing is inefficient and labour intensive. Mechanization of the various operations, in particular the use of hydraulic or continuous screw expellers or application of solvent extraction, will improve oil extraction efficiency considerably. Pre-treatment of the kernel paste with enzymes (e.g. proteases and cellulases) may also result in higher extraction rates[299]. In spite of their slightly laxative properties, mature fresh fruits are considered an important

local food[299]. They are commonly eaten in savannah regions because they ripen during the land preparation and planting season[299].The pulp has a sweet flavour[299]. The flowers are also considered an important local foods[299] They are sometimes made into fritters[299]. The reddish latex which exudes from deep cuts in the bark is used as a chewing gum[299].

[References](#) [More on Edible Uses](#)

Medicinal Uses

Plants For A Future can not take any responsibility for any adverse effects from the use of plants. Always seek advice from a professional before using a plant medicinally.

[Antibacterial](#)[Antidermatosis](#)[Antidiarrhoeal](#)[Antiinflammatory](#)[Antirheumatic](#)[Dysentery](#)[Odontalgia](#)[Ophthalmic](#)
[Skin](#)

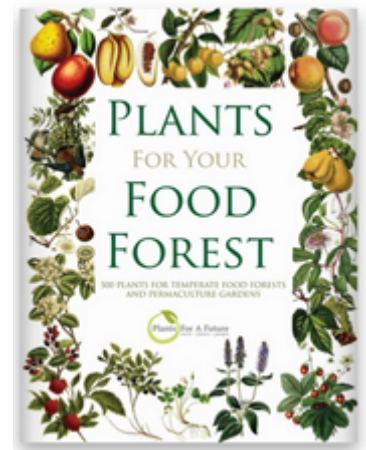
Shea butter is a suitable base for topical medicines. Its application relieves rheumatic and joint pains and heals wounds, swellings, dermatitis, bruises and other skin problems[299]. The high proportion of unsaponifiable matter, consisting of 60 - 70% triterpene alcohols, gives shea butter creams good penetrative properties[299]. Allantoin, another unsaponifiable compound, is responsible for the anti-inflammatory and healing effect on the skin[299]. The fat is used traditionally to relieve inflammation of the nostrils[299]. Clinical tests with patients suffering from rhinitis, and having moderate to severe nasal congestion, showed that shea butter may relieve nasal congestion better than conventional nasal drops[299]. The leaves are used to treat stomach-ache[299]. They are also added to vapour baths to treat headache and as an eye bath[299]. Ground roots and bark are used to treat diarrhoea, jaundice and stomach-ache[299]. Bark infusions have medicinal and antimicrobial properties, e.g. against dysentery[299]. They are applied as an eyewash to counteract spitting-cobra venom[299]. A bark decoction has been used in baths to facilitate childbirth and stimulate lactation among feeding mothers[299]

[References](#) [More on Medicinal Uses](#)

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An important new book from PFAF. It focuses on the attributes of plants suitable for food forests, what each can contribute to a food forest ecosystem, including carbon sequestration, and the kinds of foods they yield. The book suggests that community and small-scale food forests can provide a real alternative to intensive industrialised agriculture, and help to combat the many inter-related environmental crises that threaten the very future of life on Earth.

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Other Uses

CharcoalCosmeticFertilizerFilterFuelFurnitureGumHairMulchOilSoap_makingTeethWood

Agroforestry Uses: The tree regenerates well, and is traditionally favoured and protected by farmers. As a result, it has played a significant role in soil and water conservation and environmental protection in semi-arid West Africa[303]. The husks of the seeds make a good mulch and fertilizer[303]. Studies on the by-products of shea-butter processing have shown that heavy-metal ions can be removed from aqueous solutions, for example waste water, using Vitellaria seed husks[303]. The tree combines well with many cereal crops[303].

Other Uses Shea butter is a vegetable fat obtained from the seed[63 , 299]. Many cosmetic products, especially moisturizers, lotions and lipsticks, use it as a base because its high unsaponifiable matter content imparts excellent moisturizing characteristics[299]. The high proportion of unsaponifiable matter, consisting of 60 - 70% triterpene alcohols, gives shea butter creams good penetrative properties that are particularly useful in cosmetics. Allantoin, another unsaponifiable compound, is responsible for the anti-inflammatory and healing effect on the skin. It is used in toothpastes and other oral hygiene products, in shampoos, lipsticks, cosmetic lotions and creams, and other cosmetic and pharmaceutical products[299]. Low-quality shea butter, often mixed with other oils, is a base material for soap[299]. Shea butter is also very suitable for making candles because of its high melting point[299]. As a waterproofing agent, shea butter is used as daubing for earthen walls, doors and windows[299]. The black sticky residue, left after oil extraction, is used to fill cracks in walls and also as a waterproofing material[299]. Waste water from shea butter production has pesticidal properties[299]. The press cake and the husks remaining after oil extraction are potential fertilizers and fuels[299]. The leaves, soaked in water, produce a good lather for washing[299]. The reddish latex (gutta shea or red kano rubber) which exudes from deep cuts in the bark is made into glue, chewing gum and balls for children's games[299]. Musicians use it to repair drums[299]. Only unproductive and unhealthy trees are cut for timber[299]. The wood is moderately heavy and hard. It is liable to crack on drying and needs to be seasoned slowly. It is difficult to work and tends to split on

sawing, but it polishes well. It glues, nails and screws well, but pre-boring is recommended to avoid splitting. It is durable and resistant to termites. Both sapwood and heartwood are resistant to impregnation with preservatives[299]. The wood is used for poles, house posts, rafters, flooring, domestic utensils and furniture[299]. It is an excellent fuel wood, burning with great heat, and a source of charcoal[299].

Special Uses

[Carbon Farming](#) [Food Forest](#)

[References](#) [More on Other Uses](#)

Cultivation details

Management: StandardMinor Global CropStaple Crop: Oil

Shea butter tree is a plant of the drier lowland tropics, usually at elevations up to 600 metres, though it can also be cultivated at elevations up to 1,500 metres[303 , 418]. It grows best in areas where annual daytime temperatures are within the range 24 - 38°C, but can tolerate 18 - 43°C[418]. It prefers a mean annual rainfall in the range 500 - 1,000mm, but tolerates 300 - 1,800mm[418]. Grows best in a sunny position, tolerating light shade[418]. Succeeds on a variety of soils such as clay; sandy clay; sand; stony soil and laterites. It prefers colluvial slopes with moderately moist, deep soils, rich in organic matter[299]. Plants can also succeed in poor, lateritic soils[335]. Prefers a pH in the range 6 - 7, tolerating 5.5 - 8[418]. There are two main forms of the plant:- Ssp. Paradoxa grows at lower elevations, mostly at 100 - 600 metres, though can be found as high as 1,300 metres[299]. The mean annual temperature in its range is 25 - 29°C, with a mean annual rainfall of 600 - 1,400mm and 5 - 8 months dry season where the precipitation is less than 50 mm[299]. Ssp. Nilotica is found at the somewhat higher elevations of 450 - 1,600 metres where the mean annual rainfall is 900 - 1,400 mm and there are only 3 - 5 dry months[299]. Shea butter trees have been protected by farmers for many centuries in the West African savannah, particularly where cattle are scarce. Productive trees are retained when new fields are cleared, leading to areas in Sudan where more than 40% of the trees are Shea butter. Natural regeneration is favoured by fallow of at least 5 years. Shortening the fallow period leads to insufficient regeneration. In areas of cultivation, shea butter tree is found in association with annual crops, such as pearl millet, sorghum, groundnut, cotton, cassava, yams and vegetables[299]. The tree produces a taproot up to 1, occasionally 2 metres long; with shallow lateral roots that are concentrated at a depth of 10 cm and extend up to 20 metres outward from the tree; and secondary lateral roots growing downwards to the same depth as the tap root[299]. The taproot and secondary root system strongly develop during the first few years of growth. This enables the seedling to produce new shoots when the original ones are damaged by drought or fire[299]. Early stem growth is slow; branching occurs after 4 - 7 years. The tree begins flowering at the age of 10 - 25 years. Early flowers may be sterile. Maturity is reached at 20 - 45 years and the total lifespan is around 200 - 300 years[299]. Leaf fall, flowering, flushing and the onset of fruiting occur during the dry season. Leaves drop

mostly at the beginning of the dry season. Trees are rarely completely leafless, or only for relatively short periods. Flowering occurs from the beginning to the middle of the dry season. About 25% of the flowers set fruit. Fruits develop in 4 - 6 months; maturation peaks in the rainy season[299]. Productivity of the trees is variable. In a sample taken in Burkina Faso, the best 25% of the trees produced 60% of the yield, while the poorest 30% of trees produced little fruit[299]. A good tree can bear on average 15 - 30 kg fruits per year. In a good year this may be as much as 50 kg, but then only about 15 kg in the next two years[299] Although a clear production cycle is not confirmed, observations show a tendency for trees to give only 1 good harvest every 3 - 4 years[299]. Although the trees are fire tolerant, their growth and fruiting are affected by fire. Therefore, trees must be protected by ring weeding[299]. The tree is an important source of honey. Beehives placed in its branches are assured a good supply of nectar and pollen[299].

Carbon Farming

- Management: Standard Plants grow to their standard height. Harvest fruit, seeds, or other products. Non-Destructive management systems.
- Minor Global Crop These crops are already grown or traded around the world, but on a smaller scale than the global perennial staple and industrial crops, The annual value of a minor global crop is under \$1 billion US. Examples include shea, carob, Brazil nuts and fibers such as ramie and sisal.
- Staple Crop: Oil (0-15 percent protein, 16+ percent oil). Some of these are consumed whole while others are exclusively pressed for oil. Annuals include canola, poppyseed, maize, cottonseed, sunflower, peanut. Perennials include high-oil fruits, seeds, and nuts, such as olive, coconut, avocado, oil palm, shea, pecan, and macadamia. Some perennial oil crops are consumed whole as fruits and nuts, while others are exclusively pressed for oil (and some are used fresh and for oil).

References [Carbon Farming Information](#) and [Carbon Sequestration Information](#)

Temperature Converter

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Fahrenheit:



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Propagation

Seed - should not be dried, but sown as soon as possible because their viability is very short. When fresh seed is used, germination is 90 - 97% at 25 - 30°C[299]. Storing seed at 25°C for 70 days and 140 days resulted in 96% and 88% germination, respectively[299]. Seed can be planted directly in the field or in the nursery. Seed-beds are made of a mixture of organic compost and sand[299]. Seeds are planted at 1 - 5 cm depth[299]. After 1 year, seedlings are transplanted in the nursery or planted directly in the field[299]. Those grown in containers are transplanted after 1 - 2 years[299]. Vegetative propagation has only been successful in experiments. Grafting can accelerate the fruiting of the tree. In experiments in Burkina Faso, some grafted seedlings started to bear fruit

one year after grafting[299]. Latex exudation interferes with rooting of cuttings and with grafting. A 25% success rate can be achieved in grafting if the scion is soaked in water for a few hours to allow the latex to drain[299]. Marcotting has been tried with some success; growth hormones improved the success rate[299].

Other Names

If available other names are mentioned here

Anku, Bambo-tulo-iro, Bugbassami, Busabu, Carei, Careidje, Carite, Chobu, Ekumgurit, Ekunguru, Gi-wol, Kare, Karite, Kotoble, Leguelcare, Lulu, Midji, Nguin, Nku, Nkudua, Oum kouroum, Taana, Taang-kaam, Taanga, Tabi, Tongtia, Wado, Yaa, Yao, arbre a beurre, arbre à beurre, ayiokumi bati, balili, balire, bambo-tulo-iro, bambouk-buttertree, blankaidie, bugbassamu, bulanga, busabu, butirospermo, carei, carité, cere, chammal, chobu, cárei, emi, emi-egidi, galam, galam-buttertree, gi-wol, hoiti, kadainya, kadampo, kadania, kadanya, kanku, karate, kare, karedie, karehi, karite, karite-nut, karité, kedempo, kochi, kotoble, kra-nku, kwohakwe, lontingue, lotchigue, lulu, midji, ngu, nguin, njab, njabi, nku, nkudua, okwuma, ori, osisi, saambu, schibutterbaum, se, se, shi, si, se, si, sii, si iri, se-na, shea, shea butter, shea butter tree, shea tree, shea-buttertree, sheanut oil, refined, sheasmörträd, sheatree si, si/se/sii, somu, taana, taana, ta-anga, tabi, tan, tanga, tubbi, yo, yokuti, árbol montequero.

Found In

Countries where the plant has been found are listed here if the information is available

Cameroon; Congo, The Democratic Republic of the; Côte d'Ivoire; Ghana; Guinea; Nigeria; Senegal; South Sudan; Sudan; Uganda, Africa, Benin, Burkina Faso, Central Africa, Central African Republic, CAR, Côte d'Ivoire, East Africa, Ethiopia, Gambia, Guinée, Guinea-Bissau, Ivory Coast, Mali, Niger, Sierra Leone, Togo, West Africa,

Weed Potential

Right plant wrong place. We are currently updating this section. Please note that a plant may be invasive in one area but may not in your area so it's worth checking.

Conservation Status

IUCN Red List of Threatened Plants Status : Status: Vulnerable A1cd



Expert comment

Author

C.F.Gaertn.

Botanical References

Links / References

For a list of references used on this page please [go here](#)

A special thanks to [Ken Fern](#) for some of the information used on this page.

Readers comment
